10.4.1 Paper electrophoresis

Paper electrophoresis is an inexpensive method and requires only micro-quantities of protein. The apparatus consists of two troughs to accommodate a buffer through which an electric current is applied (Fig.10.9). Paper is a popular support medium as it is easy to handle, less expensive and is readily available. Paper contains 98% of cellulose. Paper electrophoresis has potential limitations. The greatest problem is the thickness and large pore size of the paper. The separation of proteins by paper electrophoresis takes longer time which limits its use.

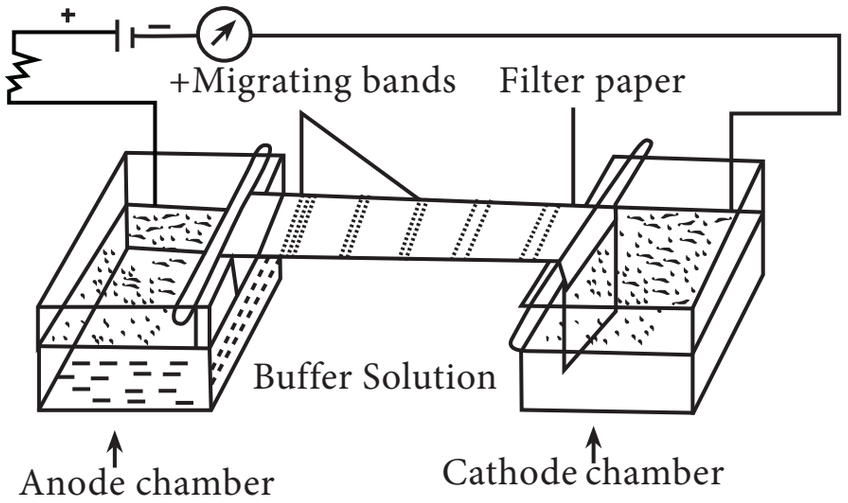


Figure 10.9 Paper electrophoresis

10.4.2 Gel Electrophoresis

i. Polyacrylamide Gel Electrophoresis

Polyacrylamide gel is prepared from acrylamide and bis-acrylamide in a suitable buffer. Polymerization of Acrylamide and bisacrylamide is achieved by a free radical reaction promoted by N,N,N',N'tetramethylethylenediamine (TEMED). This free radical process is initiated by Ammonium per sulfate (APS) used in gel. Acrylamide and bisacrylamide monomers are weak neurotoxin whereas, the polymerised polyacrylamide is non-toxic. While handling acrylamide solutions, care should be exercised and spectacles, gloves and mask should be worn.

ii. Sodium dodecyl Sulphate (SDS) polyacrylamide gel electrophoresis

Sodium dodecyl Sulphate Polyacrylamide Gel Electrophoresis(SDS-PAGE) is an electrophoretic technique very commonly used in Biochemistry, Molecular biology and forensic science. This technique was first described by Laemmli in the year 1970 and till now dominates in scientific research.

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